Deep Research of Copilot

Mechanistic Estimation of Deep Third-Cumulant Content in Random ReLU Networks

Executive Summary
The analysis concludes that the new parity and sensitivity measurements strongly constrain possible explanations for leaderboard-leading methods. The expensive object is unlikely to be the old sources themselves. Instead, the burden appears to come from carrying a source-labeled representation of the odd third-order sector. Any successful formulation must preserve quenched, realization-specific odd structure while avoiding per-source dense Hadamard contractions.

Q1(c) Slice Recursion

Carry only D21 and D3. Approximate transport with D21' ≈ (W■W) D21 W^T plus D3 corrections. Cost about 2 units per layer, 32 units total. First falsifier: measure residual energy in exact next-layer D21 unexplained by (D21, D3, C) as a function of source age.

Q1(a) Hermite Field Representation

Replace source-wise storage with one merged degree-3 chaos field. Carry a global matrix B representing the H2⊗H1 sector. Cost roughly 3 units per layer. This preserves aggregate odd structure while discarding source genealogy. Falsifier: project the exact summed old-source contribution into the degree-3 Hermite basis and measure captured variance.

Q1(d) Exact Low-Cost Identity

No convincing exact O(n r^2) identity is apparent. The observed nearly full-rank Hadamard spectra indicate that Khatri-Rao rank inflation is real and numerically realized. This argues against further exact factor-space compression.

Shared Propagator Hypothesis

A single common propagator with per-source scalars is inconsistent with the nearly flat birth-ablation sensitivities. Falsifier: compute principal angles among old-source propagator subspaces.

Q2: Leader Hypotheses

1. Merged degree-3 field (most plausible). 2. Direct recursion on D21. 3. Effective even-order renormalization mimicking old-source effects. Distinguish using scaling with source count and predictability tests.

Q3: Most Informative Measurement

Measure predictability of next-layer D21 from current (C, D21, D3). If residual unexplained energy stays high, no low-order closure exists and leaders must carry another hidden odd field. If residual drops below about 2% for old sources, a cheap recursion becomes credible.

Key Conclusions

Derived: old-source mean-field averaging is ruled out by parity. Conjecture: leaders changed representation rather than arithmetic. Most likely they carry a merged odd third-order field whose state size scales like O(n^2), not O(number of sources × n^2).